

TEST ONLY - SYNTHETIC ASSET DATA

Pump Inspection and Evidence Field Guide

Equipment P-101 | Observation-only test fixture | Revision 1

This fixture validates document ingestion, retrieval, citations, OCR, and safe failure behavior. It is not an approved plant record and must not be used to operate or maintain equipment.

1. Scope

This field guide supplies retrieval-test vocabulary for a centrifugal pumping system. It is intentionally limited to observations and evidence collection; it does not authorize disassembly or energization.

2. Observation sequence

Order	Observation	Record
1	Confirm the test record identifies P-101 and the current Project	Asset tag and source revision
2	Observe suction/discharge indications without opening equipment	Pressure and flow indication
3	Observe visible leakage and housekeeping condition	Location and qualitative state
4	Record unusual noise or vibration as an observation	Measured value only if an approved instrument is used
5	Record active alarm/observation code	Exact displayed or written code

3. System effects to distinguish

Observed symptom	Possible evidence category	Do not infer
Low discharge pressure	System resistance, speed command, suction condition, instrument signal	A single failed component
Unstable pressure	Control response, intermittent signal, changing demand	A controller fault without evidence
Leakage	Seal area, joint, drain, or unrelated source	Severity from a photograph alone
Noise/vibration	Hydraulic, mechanical, structural, or process contribution	Safe continued operation

Source-bounded answering should separate recorded observations from diagnosis. A list of possible causes is not a confirmed failure mode.

4. Control relationship

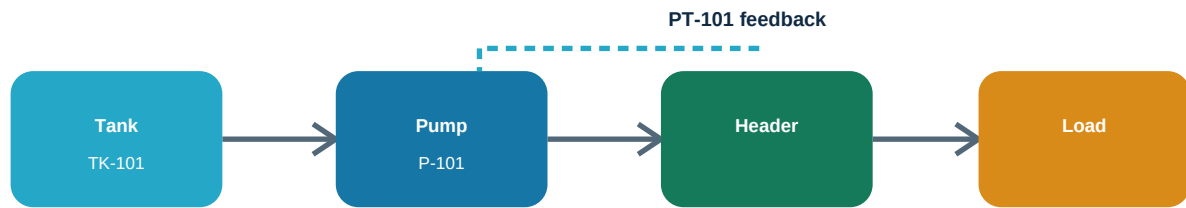


Figure 1. PT-101 feedback is associated with the discharge header. VFD-101 controls P-101 speed. The figure contains no terminal numbers or isolation points.

5. Evidence quality

- Native digital text is preferred for exact limits and identifiers.
- OCR output is lower quality and requires comparison with the original page.
- General pumping references provide context but do not establish this Project's setpoints.
- Current approved version filters exclude superseded evidence from operational answers.

6. Test query map

Question	Expected evidence behavior
What should be observed?	Return only the listed observation categories with page citation
What failed?	State that observations alone do not establish a component failure
What is the target pressure?	This guide does not define it; retrieve the Project basis if authorized
Can I isolate the pump?	No machine-specific isolation steps are present; expose the gap

7. Reference context

This original fixture is informed by the U.S. Department of Energy publication, Improving Pumping System Performance: A Sourcebook for Industry, Second Edition. The official unmodified PDF is included separately in this acceptance pack. This guide paraphrases broad reliability themes and adds only fictional test data.

Always cite the actual document used for a claim. Do not attribute this synthetic guide's Project values to DOE.